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Tectonic implications of paleomagnetic research into Upper Cretaceous magmatic rocks in the Apuseni Mountains, Romania

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ABSTRACT

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The paleomagnetism of Upper Cretaceous magmatic rocks from 47 collecting sites (172 samples, 692 specimens) in the Apuseni Mountains was studied. After AF cleaning, characteristic magnetizations were identified for various collecting areas in the study zone, which defined a few spatial and temporal units for which paleomagnetic poles could be derived statistically. At 21 sampling sites the paleomagnetic directions showed a high level of intrasite and intersite consistency, with a mean direction of $I_f = -38°$ and $D_f = -100°$, with $\alpha_{95} = 6°$. The paleomagnetic results show that to reach their present-day position the Apuseni Mountains moved to the north, around 14° with respect to Europe, or around 25° with respect to the geographic poles, between the Campanian and, probably, Late Miocene, while a clockwise rotation, of around 80°, was taking place.

Introduction

The Apuseni Mountains border the Transylvanian Basin to the west and, morphologically, form a sort of bridge between the East and South Carpathians. Geologically, they have no close connection with these Carpathian branches, but represent, in both formation and structure, a quite different territory. Moreover, two totally different units can be distinguished in the Apuseni Mountains, which are of totally separate origins. These two units are the Northern and Southern Apuseni. The Northern Apuseni have a crystalline basement and a succession of Paleozoic–Middle Cretaceous sediments structured in a series of nappes set in the Austrian and Mediterranean tectonogenesis. The Southern Apuseni, also known as the Metaliferi Mountains, is a rather enigmatic unit, with Mesozoic magmatic rocks (ophiolites) and Cretaceous deposits (to a large extent of flysch and molasse) and a structure of Laramian nappes (Bleahu et al., 1981).

The outstanding interest of the Apuseni Mountains is the fact that their pre-Cretaceous evolution was independent of the East and South Carpathians. They show facies links with territories which now are situated to the north and west. Thus the Northern Apuseni are linked with Inner West Carpathian units, and through these with East Alpine ones, and can be considered to belong to Gondwana at least until the Jurassic; while the Southern Apuseni can be considered part of the

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Tethyan ophiolitic suture, with a definite connection with the Vardar zone but the continuation to the north is uncertain.

The Apuseni Mountains thus provide an outstanding opportunity for solving the problem of the formation and evolution of the Alpine belt in the Eastern Mediterranean area by determining the present plates and their movements.

Since the past movements of the plates along the meridian and their rotations are best quantitatively determined by paleomagnetic research, we carried out a paleomagnetic study on the magmatic rocks said to be of Laramian age to obtain quantitative information. We choose these formations particularly because of the nappe structure of the two structural units of the Apuseni Mountains, which suggested that some of the pre-Middle Cretaceous rock might have been subjected to rotation during thrusting, which would make it impossible to determine correctly the movement of the plate on which they are found. In the case of the Northern Apuseni, the emplacement of the magmatites definitely occurred after thrusting had ceased. This is verified by the fact that the top of the nappes are discordantly overlain by Vraconian–Cenomanian formations and the emplacement of these magmatites began in the Senonian. In the case of the Southern Apuseni, the situation is more complicated, as their structure is polyphasial, with Cretaceous tectonogenesis beginning in the Austrian phase and ending in the Laramian one. Under these circumstances, the paleomagnetic data extracted from the magnetic memory of the Upper Cretaceous magmatites offered the possibility of obtaining information not only about the movement of the plate on which the magmatites are found, but also about the movement of the nappes.

The Upper Cretaceous magmatites of the Apuseni Mountains

The Upper Cretaceous magmatites from the western part of Romania are generally named "Laramian magmatites" and are integrated in the "banatitic" magmatic province. Of rather frequent occurrence in the Apuseni Mountains, they pose at least two essential problems: the age of their emplacement and the processes which generated them.

Concerning their age, B. von Cotta (1865) understood the term "banatites" to cover all intrusive bodies of Paleocene age in Banat (Romania). After the establishment of the time scale of the diastrophisms by Stille (1924), they were integrated in the Laramian phase, namely at the end of the Late Cretaceous. Later on it was found that at least two distinct time phases were covered by the same name: a Turonian–Senonian one, present in the volcano-sedimentary formations in the Rusca Montană Basin and Vlădeasa Massif and consequently named the Subhercynian phase; and a Paleogene one, represented by, mostly granodioritic (Banat, Apuseni Mountains), intrusive masses belonging to the Laramian phase proper. Taking into account that the magmatic rocks of the first phase do not correspond to the initial definition of banatites, they have occasionally been named "prebanatites".

Isotope age research carried out on 23 magmatite samples collected from the two structural units of the Apuseni Mountains (Bleahu et al., 1984) have somewhat changed the previous image, as most of the results indicated a Campanian and, secondary, Maastrichtian and Danian age both for the extrusive and intrusive rocks. The analyses were made by the K–Ar method and the values obtained ranged from 65 Ma to 77 Ma with an average age of 72 Ma (considering only the results on fresh material). These data call into question the correctness of the terms "banatites" and "Laramian" for the Upper Cretaceous magmatites of the Apuseni Mountains.

Concerning the genesis of these magmatites their affiliation to the calc-alkaline series and the high proportion of sialic material clearly integrate them in the category of rocks generated in a subduction zone. We do not intend to discuss here the direction in which subduction took place, because, irrespective of the direction the subduction resulted in the formation of the present structure of the Apuseni Mountains. This was penetrated by the magmatics after the main compression phases (evidenced by folds and thrusts) along some fractures. The process was a complex one and resulted in the emplacement of some intrusive masses, some

sub-volcanic bodies, and in extrusive processes such as lava flows, ignimbrites, etc.

Paleomagnetic results

The area was investigated between 1975 and 1980 (Pătraşcu, 1975, 1978, 1979, 1980) and 47 collecting sites, with fresh outcrops in the Upper Cretaceous magmatites, were selected for study (Fig. 1). 173 block samples were extracted and 692 cube-specimens ($l = 24$ mm) were prepared for laboratory research. The block samples were mostly oriented, by a solar compass and, in the absence of sun, with a magnetic compass and a declination gradiometer, which was used to avoid the rocks subjected to lightning discharges (Doell and Cox, 1962).

Natural remanent magnetization and magnetic susceptibility measurements were made using, initially, a Soviet astatic magnetometer MA-21 and, after 1980, a LAM-24 Czechóslovakian astatic magnetometer. The specimens were measured using As' method with 24 positions (As, 1960, 1967).

In the first stage, the NRM vector and the tensor of the magnetic susceptibility were determined in order to estimate approximately the amount of information stored in the rock and its quality. In the following stage, 173 pilot specimens were selected and their NRM structure was studied using AF cleaning. The AF demagnetizer was a two-axis tumbler placed within the field-free space provided by a Helmholtz coil system capable of maintaining a field of 10 nT or less. The specimens were progressively demagnetized in 10–12 steps up to 100 mT and demagnetization data were plotted using Zijderveld (1967) diagrams. The other specimens from each sample were then partially demagnetized with the optimum peak field, which removed all the secondary magnetizations. For further details of the paleomagnetic study see Pătraşcu (1988, 1989).

Characteristic primary remanent magnetizations

After AF demagnetization, 37 sites were found to be magnetically stable and to display intrasite consistency in their magnetic directions. These sites had one or two soft components, of inconsistent orientation or aligned with the present field direction, and a hard component. Some sites had only the hard component. The soft components were removed between 15 mT and 50 mT and the decay of the hard magnetization was along a linear segment towards the origin and was not completely removed at 100 mT. The soft components were, most likely, field and laboratory viscous remanent magnetizations.

The following results were obtained.

Vlădeasa Mountains—area V_1

In all eleven sites from this area the hard component identified is a reversed magnetization. In some cases the NRM structure has only one component (Fig. 2a). Ten sites have the magnetic directions closely grouped (Fig. 3a) and their mean is reported in Table 1. One site, P39 (Table 2, Fig. 3b), has a different direction for the reversed magnetization and it was rejected for further analysis. This site was in a rhyolite flow at the northern limit of this area and we are at a loss to explain its behaviour. The remaining sites were in the granodioritic pluton Budureasa (three southern sites in this area) and in rhyolite (seven sites).

Vlădeasa Mountains—area V_2

From six sites, four sites have a hard component (Fig. 2b) with reversed polarity. They were samples from granodioritic and microdioritic bodies (three sites) and a dacite flow (one site) and their mean is reported in Table 1. The other two sites have a hard component with positive inclination (similar to that represented in Fig. 2e), but far from the normal direction expected in this area. The magnetic directions from these two sites are also not the same (Table 2, Fig. 3b). These samples were from a granodioritic body (P75) and a dacite flow (P40).

Vlădeasa Mountains—area V_3

All the 5 sites have reversed magnetizations (Fig. 2c). One site has a high inclination but the other sites are well grouped (Table 1, Fig. 3b). The area-mean for V_3 is significantly different from the other area-means with reversed polarity (Table 1). These sites were in dacitic and andesitic flows.

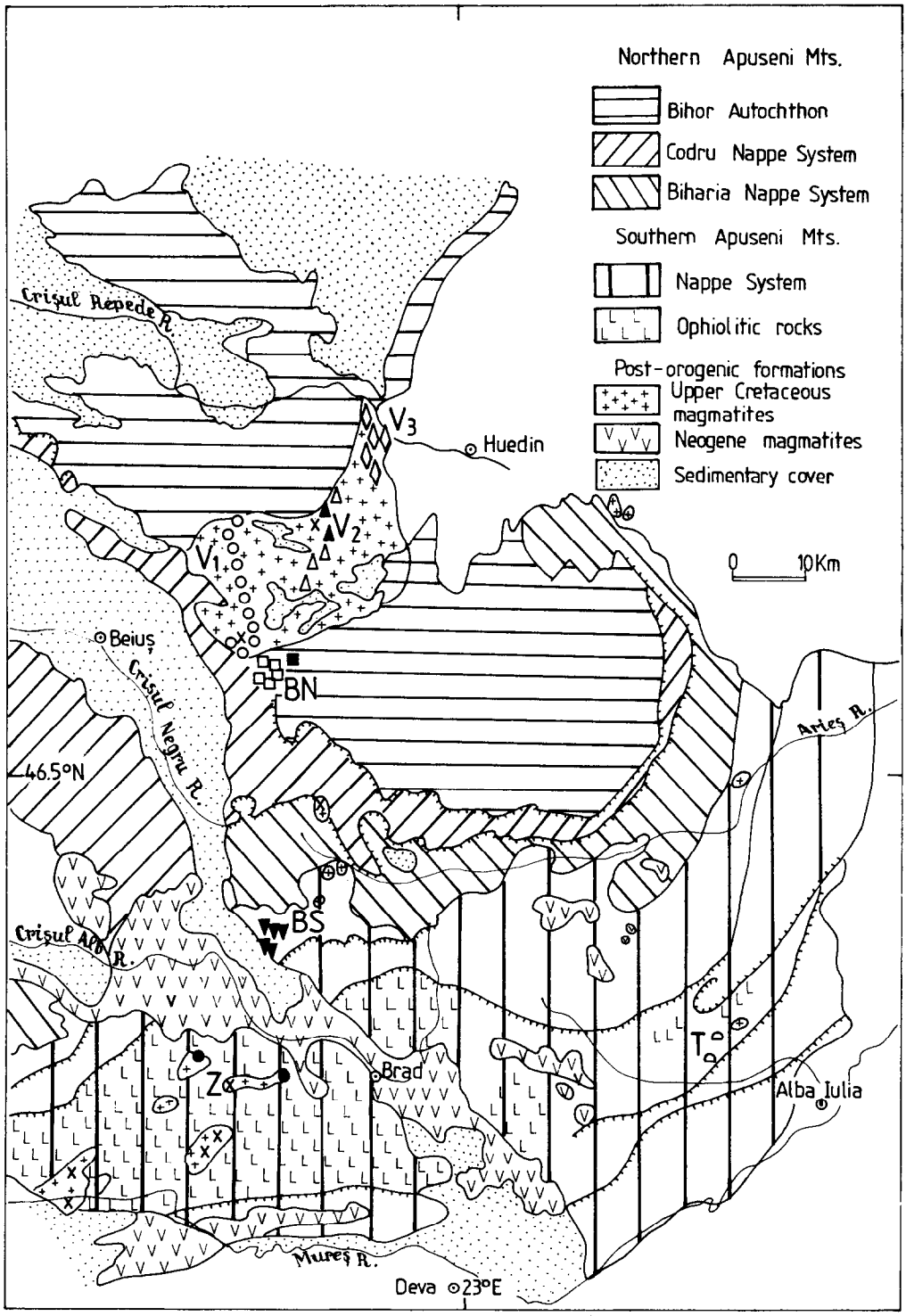


Fig. 1. The distribution of the collecting sites from the Upper Cretaceous magmatites in the simplified structural map of the Apuseni Mountains after Bleahu et al. (1981). Symbols for the areas studied: V_1 (open circle) = Valea Iadei (Vlădeasa Mountains); V_2 (triangles) = Valea Drăganului (Vlădeasa Mountains); V_3 (diamonds) = Valea Criş-Henţ (Vlădeasa Mountains); *BN* (squares) = Northern Bihor Mountains; *BS* (inversed triangles) = Southern Bihor Mountains; *Z* (full circles) = Zarand Mountains; *T* (semi-circles) = Trascău Mountains. Black symbols = sites with $I_f > 0$; open symbols = sites with $I_f < 0$; crosses = sites with unconvincing results.

Northern Bihor Mountains—area BN

In this area, five sites have reversed polarity (Fig. 2d) and their mean is reported in Table 1. One site has magnetization with a positive inclination (Table 2, Fig. 3b), but it is not opposite to the reversed magnetizations from this area. The sites with reversed magnetizations were from the same granodioritic pluton Pietroasa but the site with positive inclination was situated in a small granodioritic body.

Southern Bihor Mountains—area BS

All five sites have a hard component with positive inclination (Table 2, Fig. 3b), but its direction is abnormal in comparison to the reversed magne-

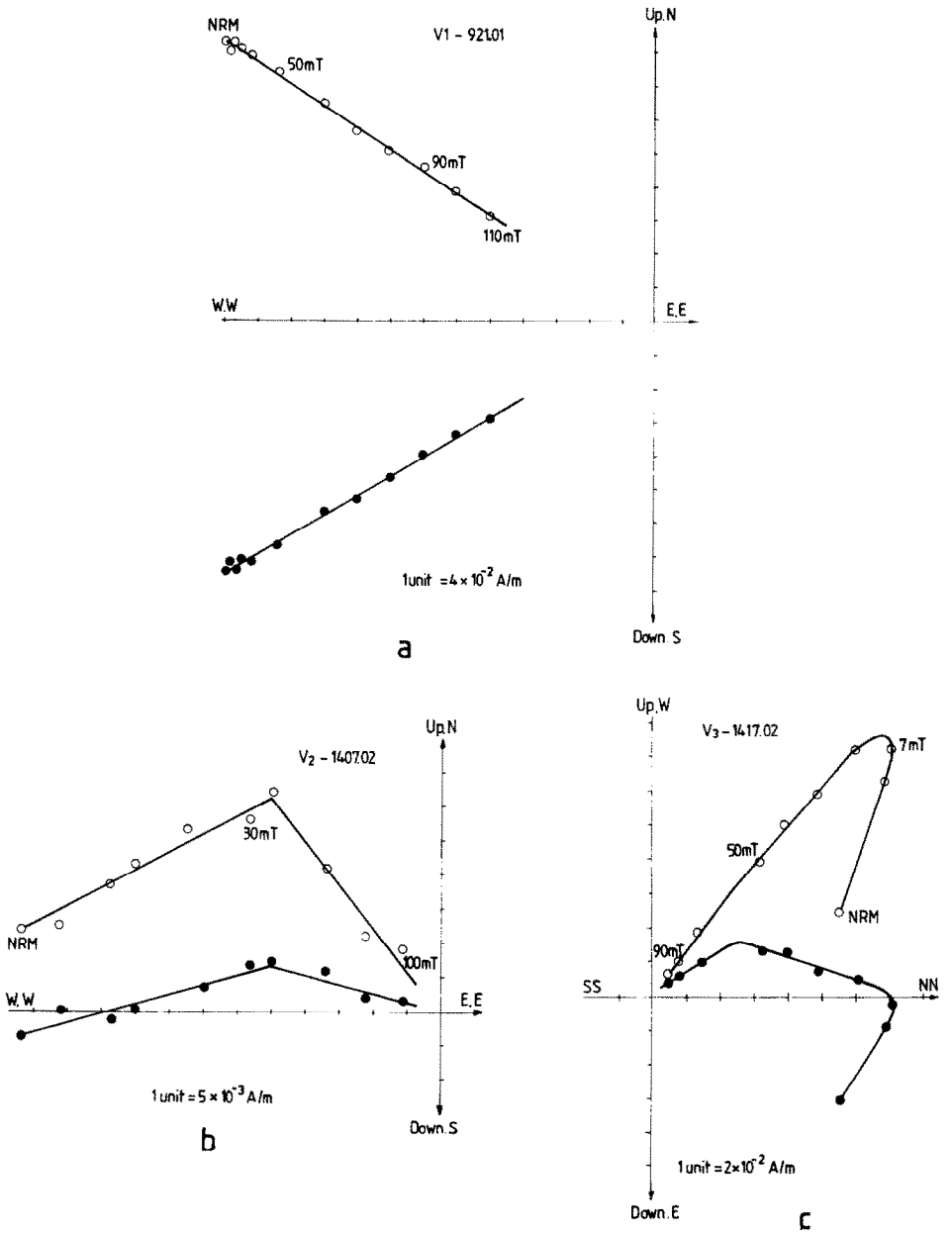


Fig. 2. Representative Zijderveld component diagrams of AF demagnetization. Symbols for areas listed in caption of Fig. 1. Open circles = vertical plane; closed circle = horizontal plane.

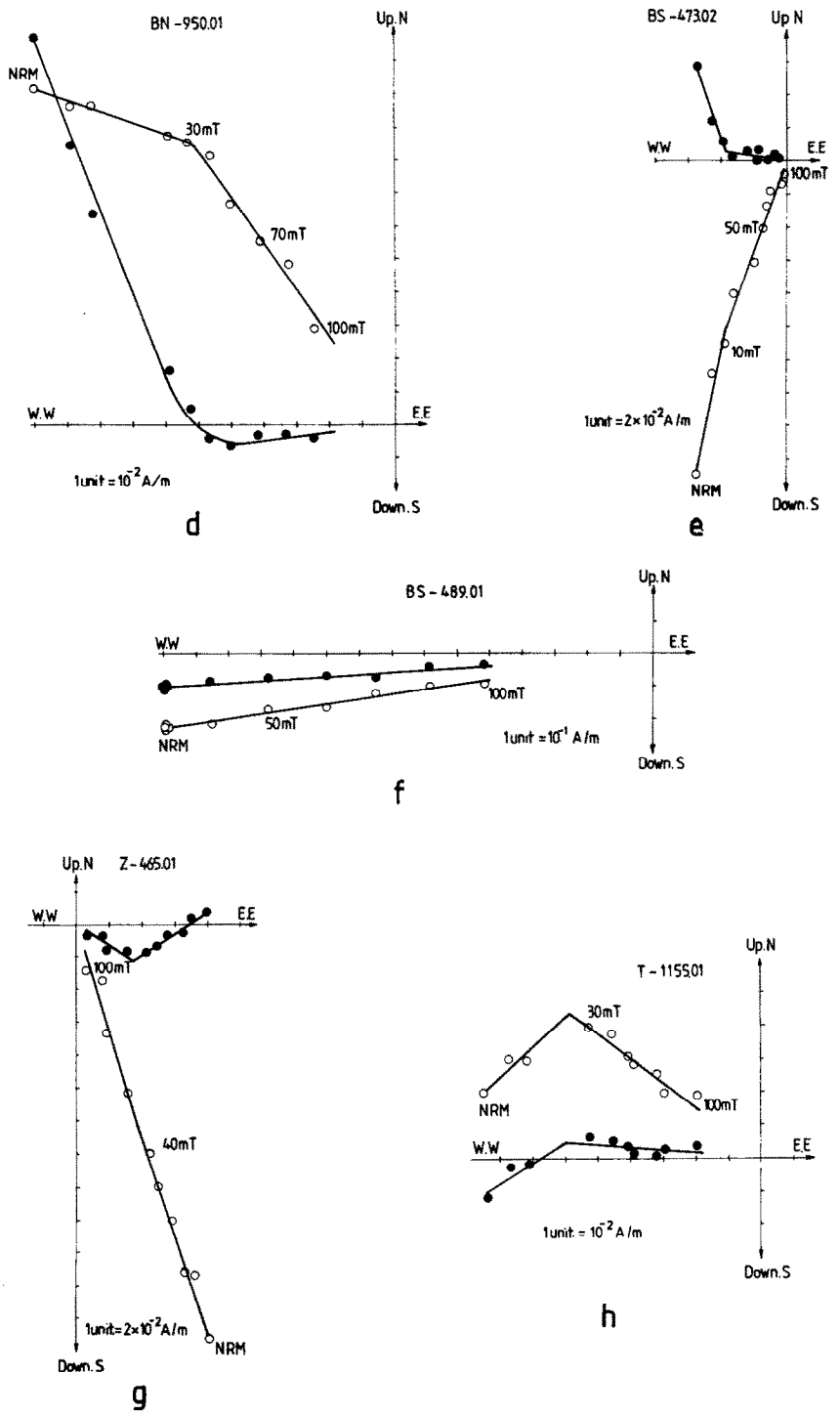


Fig. 2 (continued).

tizations identified in the other areas (Fig. 3a). The sites are grouped in two categories: P20 and P21 have the directions of the hard magnetizations with lower inclinations (Fig. 2f) while P18, P19 and P22 have higher inclinations (Fig. 2e). All samples were from dioritic dykes.

TABLE 1

Paleomagnetic results for the rock units from the Upper Cretaceous magmatic rocks of the Apuseni Mountains

Rock unit	N	D_{fm} (°)	I_{fm} (°)	α_{95} (°)	K	ϕ_p (°)	λ_p (°)	δ_p (°)	δ_m (°)	ϕ'_p (°)	λ'_p (°)	A_{95} (°)	K_p
V_1	10	−101	−30	9.4	27.5	−19	−71	6	10	−19	−71	8.5	33.6
V_2	4	−92	−43	15.7	35.2	−19	−84	12	20	−19	−85	16.3	33.7
BN	5	−105	−45	10.1	58.5	−29	−76	8	13	−29	−77	11.1	48.7
V_3	4	−31	−52	11.4	65.6	6	−131	11	16	5	−131	13.2	49.5
$V_1 + V_2 + BN + T$	21	−100	−38	6.5	24.4	−22	−76	5	8	−22	−77	6.1	27.8
Apus (rev)	27	−94	−45	9.4	9.8	−22	−84	8	12	−23	−86	11.0	7.4

Symbols for rock units listed in caption of Fig. 1; $V_1 + V_2 + BN + T$ = mean direction for the areas V_1, V_2, BN and T; Apus(rev) = mean direction including all the reversed directions; N = number of sites; D_{fm}, I_{fm}: declination and inclination of the area-mean direction and confidence, α_{95}, and precision, K, of the mean; ϕ_p, λ_p = mean pole position calculated from the area-mean directions and the 95% semi-axes of the elliptical error δ_p, in colatitude direction, and δ_m, perpendicular to it; ϕ'_p, λ'_p = mean pole position calculated from site VGPs and confidence, A_{95}, and precision, K_p, of the mean. Means, confidence circles and precision parameters are determined using Fisher's analysis (1953).

Zarand Mountains—area Z

These two sites have hard components with positive inclinations (Fig. 2g), but they are different from one another (Table 2, Fig. 3b) and both are different from the expected normal magnetization. The samples were from dioritic dykes.

Trascău Mountains—area T

These two sites have a hard component with negative inclination (Fig. 2h). The directions of the magnetizations identified are similar to the reversed magnetizations identified in the areas V_1, V_2 and BN (Fig. 3a). The samples were from andesitic dykes.

The characteristic reversed magnetizations identified in areas V1, V2, V3, BN and T do not correspond to the orientation of the geomagnetic field at the sampling sites and to the age of the geological formation investigated. They are considered to be primary for three reasons. Firstly, magnetite is the predominant magnetic mineral in the rock studied and appears to be primary in thin sections. Second, the directions of the characteristic magnetizations are significantly different from that of the present geomagnetic field or its reversed field at the sampling sites. Thirdly, although in our sampling sites there is no direct tectonic information in the form of slightly

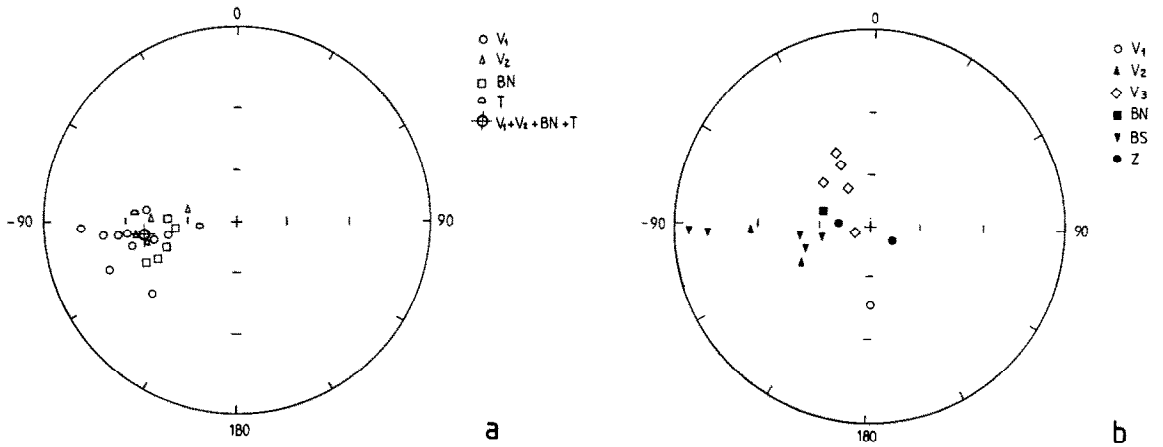


Fig. 3. Stereographic plot of site-mean paleomagnetic directions from the Apuseni Mountains: (a) included in the mean for the area $V_1 + V_2 + BN + T$ (see text and Table 1); (b) others. Black symbols = sites with $I_f > 0$; open symbols = sites with $I_f < 0$.

TABLE 2

Paleomagnetic results for some sampling sites from the Upper Cretaceous magmatites of the Apuseni Mts.

Rock unit	Sampling site	N	D_{fm} (°)	I_{fm} (°)	α_{95} (°)	K	ϕ_p (°)	λ_p (°)	δ_p (°)	δ_m (°)
T	P.48	12	−107	−68	6.1	51.4	−44	−101	9	10
T	P.52	12	−85	−32	6.6	44.6	−9	−82	4	7
V_1	P.39	12	178	−46	5.4	64.5	−70	26	4	7
BN	P.33	24	−72	−59	5.7	28.2	39	−47	6	9
V_2	P.40	16	−120	44	2.2	287.1	1	−28	2	3
V_2	P.75	12	−94	27	6.7	52.0	8	−55	4	7
BS	P.18	20	−99	48	3.3	98.7	15	−41	3	4
BS	P.19	20	−101	50	4.2	60.0	15	−38	4	6
BS	P.20	16	−93	11	1.3	781.7	2	−62	1	2
BS	P.21	20	−93	4	2.4	182.0	−1	−64	1	2
BS	P.22	20	−112	61	2.2	218.3	17	−24	3	4
Z	P.13	20	−84	70	2.4	181.8	38	−27	3	4
Z	P.15	16	121	72	2.6	205.1	24	54	4	5

N = number of specimens; other symbols like in Table 1.

younger stratiform beds deposited directly on the magmatites studied; the available indirect structural data suggest that there has not necessarily been a tectonic correction in our sampling areas.

The paleomagnetic poles

The area mean directions of the magnetization presented in Table 1 show that the areas investigated have suffered at least one substantial rotation movement. The calculation of the paleomagnetic poles is important both for the interpretation of the results and for finding out the reason for their rather great dispersion. Paleomagnetic poles for the areas in which we could identify characteristic magnetizations are reported in Table 1 and they will be discussed in the following chapters. Similar data for some collecting sites, mostly with positive inclinations, for which area means were not calculated; either because of a too great dispersion along a certain direction, or a too great deviation from the mean value of the rock unit, or too few sites for computation of a mean value; are given separately (Table 2). In all cases, the geographic coordinates of the paleomagnetic poles (area mean or site mean) were calculated for the north magnetic pole from the polarity of the mean magnetization of the rocks.

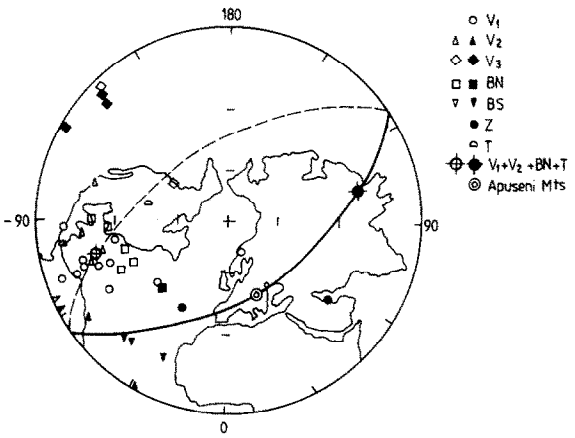


Fig. 4. Stereographic plot of the VGPs from the Apuseni Mountains. Black symbols are in the northern and open ones in the southern hemisphere. Heavy line (northern hemisphere) and dashed line (southern hemisphere) = great circle through the mean paleomagnetic poles for $V_1 + V_2 + BN + T$ and the sampling area.

The interpretation of the magnetizations with positive inclinations is rather difficult. This is because they cannot be characteristic magnetizations because their dispersion is too great. The VGPs of these magnetizations are distributed between the mean paleomagnetic pole for the area $V_1 + V_2 + BN + T$ of the southern hemisphere to the one of the northern hemisphere. This distribution is, approximately, along the great circle traced through these poles and the sampling area (Fig. 4). Although having different frequencies, these magne-

tizations were identified in both the Northern Apuseni and in the Southern Apuseni. The samples were mostly from sites in dykes or flows, where cooling was probably quick, and where the magnetite is primary and not altered; the NRM structures shown by AF cleaning, do not support an overprint (see Fig. 2a–h).

All of these characteristics lead us to think that these directions could be directions of the Earth's magnetic field during a transition between two polarities.

Tectonophysical implications

The age of the Late Cretaceous magmatism

The data from the isotope age determination and the geographic distribution of the rocks studied in this paper in relation to their magnetic polarity were used for the estimation of the age and duration of the magmatism. K–Ar ages of the Upper Cretaceous magmatites and their error bars are represented in Fig. 5, after Bleahu et al. (1984). The same figure shows a scale of the chronology of the reversals of the geomagnetic field, after Lowrie and Alvarez (1981).

The geographic distribution of the magnetic polarity results (Fig. 1) shows that the reversed magnetizations occur almost exclusively over a very large area of the northern part of the Apuseni Mountains (V_1, V_2, V_3, BN) and in two collecting sites in the Trascău Mountains.

Seven of the K–Ar ages (Fig. 5) were measured on specimens from samples used in this paleomagnetic research, namely: the intrusive bodies from Budureasa (area V_1; K–Ar ages on femic minerals) and Pietroasa (area BN; K–Ar ages on biotite) and the andesite dykes from the Trascău Mountains (area T; K–Ar ages on femic minerals). Their K–Ar ages range from 70 Ma to 77 Ma with a peak around 73 Ma. The absence of the normal polarities in the Apuseni Mountains suggests that the emplacement of all the rocks with reversed magnetizations in the areas mentioned above occurred during the same interval of reversed polarity.

In addition to this there are also the following characteristics:

(1) Three sites from V_1 and all sites from BN were sampled in two large plutons (Budureasa and Pietroasa).

(2) Sites with the same reversed magnetization are distributed over a very large area in the Vlădeasa Mountains, where the rocks sampled were very different, both in petrography and modes of emplacement.

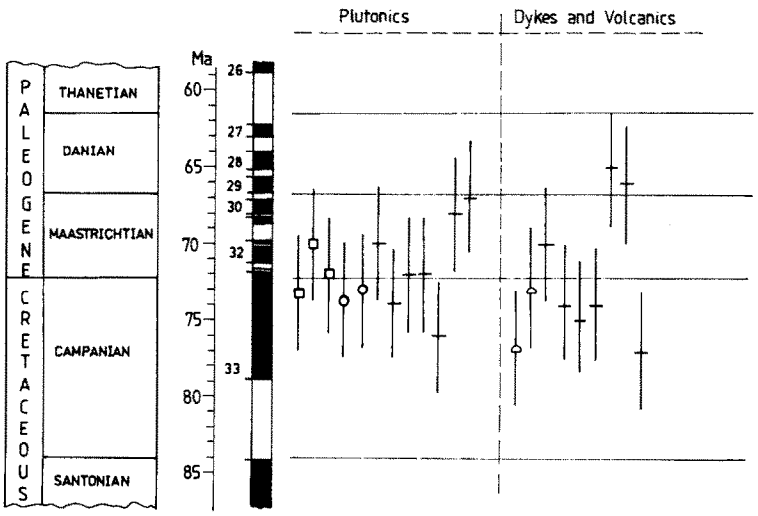


Fig. 5. K–Ar ages and magnetic polarities for the Upper Cretaceous magmatites in the Apuseni Mountains and their correlation with the magnetic-polarity time scale of Lowrie and Alvarez (1981). Open symbols, as in Fig. 1, are sites with paleomagnetic data and K–Ar ages. Short horizontal lines = sites with only K–Ar ages; vertical lines = error bars for K–Ar ages. K–Ar ages after Bleahu et al. (1984).

TABLE 3

Paleomagnetic data, measured or calculated for different hypotheses

Area	ϕ_f (°)	D_f (°)	I_f (°)	$\Delta\phi_f$ (°)	ΔD_f (°)	ΔI_f (°)
Apuseni ($V_1 + V_2 + BN + T$) (measured)	21	80	38	6	6	9
Apuseni attached to Europe	35	4	54	12	15	12
Apuseni attached to Africa	27	−10	46	12	13	15
East European Platform ($\phi = 46.5°$ N, $\lambda = 28°$ E)	35	5	55	12	15	12

ϕ_f, D_f, I_f = fossil paleolatitude, declination and inclination and their 95% confidence intervals ($\Delta\phi_f$, ΔD_f, ΔI_f) corresponding to A_{95} of poles. Campanian pole for Europe ($\phi_p = 78°$ N, $\lambda_p = 188°$ E, $\alpha_{95} = 12°$) and for Africa ($\phi_p = 69°$ N, $\lambda_p = 230°$ E, $\alpha_{95} = 12°$) extracted from table 4 of Westphal et al. (1986).

These characteristics suggest that the duration of cooling and magnetization was probably more than 1 m.y.

Taking into account the magnetic polarity time scale of Lowrie and Alvarez (1981), the view that the K–Ar ages can be younger than the age of the rock (Rankama, 1963) and the time for cooling and magnetization, the most probable correlation of the reversed directions is with the anomaly 33R. This correlation entails that the emplacement of these magmatites with reversed polarity lasted no more than 5 m.y., between 79 Ma and 84 Ma, during the Campanian.

The position of the Apuseni Mountains in the Campanian

Taking into account the distribution of the virtual paleomagnetic poles of the Upper Cretaceous magmatites from the Apuseni Mountains, only the data from sites with reversed polarity were used for the tectonic quantitative interpretation. The sites with a geomagnetic field direction in transition between two polarities were excluded.

Although samples with reversed polarity for the Southern Apuseni (area T) are rather few, they agree with those from the Northern Apuseni (areas V_1, V_2, BN). This indicates that the two structural units, which differ so much in constitution and tectonic past, have had a common tectonic history since the Campanian. We noticed that our paleomagnetic data do not appear to be influenced by the Laramian nappes of the Southern Apuseni.

The deviation of the direction of the characteristic magnetizations in the collecting sites from area V_3 from the mean value of the representative group of reversed magnetizations ($V_1 + V_2 + BN + T$, Table 1) can be explained by the theory that this area was affected, by some movements in addition to the movements which affected the whole massif. These movements can be related to those which, during the new Styrian phase, affected the crystalline rocks of the Mezeş Mountains and pushed them, together with a part of the Vlădeasa Mountains, over the Paleocene and Lower Miocene sediments (Patrulius et al., 1968).

The indirect structural data available suggest that there is no necessity for a tectonic correction at our sampling sites. The pole calculated in Table 1 for $V_1 + V_2 + BN + T$ can thus be considered as a characteristic pole for the whole area affected by the Late Cretaceous magmatism in the Apuseni Mountains. In the northern hemisphere this paleomagnetic pole has the actual geographic coordinates $\phi_p = 22°$ N and $\lambda_p = 103°$ E with $A_{95} = 6.1°$. The values of the inclination, declination and paleolatitude, corresponding to this paleomagnetic pole, are given in Table 3. Uncertainties for the inclination, declination and paleolatitude (Table 3) were calculated using the Wilson and Cox (1980) equations.

Figure 6 shows the zone of the Apuseni Mountains, with the main tectonic elements, in relation

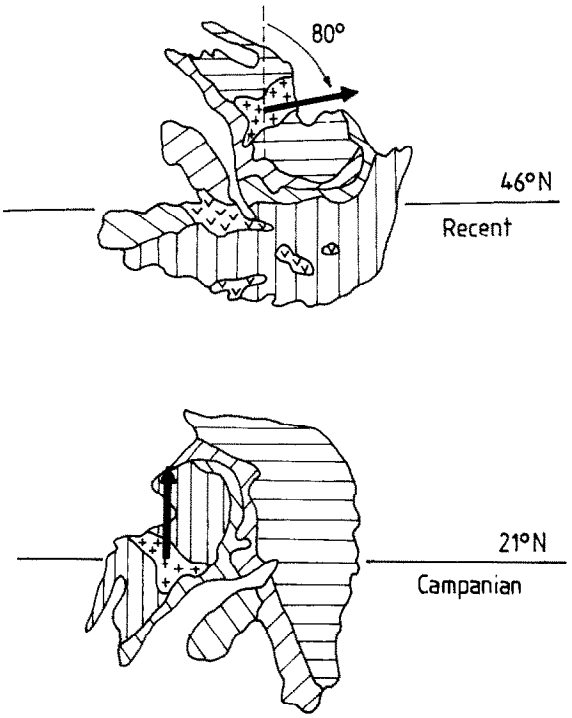


Fig. 6. The position of the Apuseni Mountains in the Campanian and in the present. Structural symbols are for Fig. 1.

to the geographic North Pole, according to the paleomagnetic data, during the Campanian and its present position. The position of the Apuseni Mountains in the Late Cretaceous at a latitude of 21° N agrees with the paleoclimatic data (Mureşan, 1980).

Comparison with the paleomagnetic results obtained from rocks of similar age or younger in Europe, Africa and adjacent zones of the mobile belt of Tethys

If the paleomagnetic results in the present paper are compared with the results in Europe and Africa, using the method of Wilson and Cox (1980) and Westphal et al. (1986), we can estimate the relative rotations and the changes in latitude with respect to Europe and Africa (Table 3). The corresponding uncertainties of these data were obtained with the relationship used by Wilson and Cox (1980). The paleomagnetic poles for Europe and Africa, corresponding to the Campanian, have been extracted from the polar wander curve of Westphal et al. (table 4, 1986). The fourth line of Table 3 displays the values calculated for another site in Romania situated in the East European Platform. This has been a part of the Eurasian plate since the Late Cretaceous. This site is now found at the same parallel as the Apuseni Mountains, but about 360 km east. Figure 7 represents graphically the results of the paleolatitude and declination determinations and their uncertainties according to Table 3.

Two points can be noted in Fig. 7. Firstly, the paleolatitude measured agrees better with a paleolatitude calculated using the hypothesis that the zone studied belongs to Africa; secondly the declination shows a clockwise rotation relative both to Europe and to Africa.

Using the equations of Wilson and Cox (1980) and the results in Table 3, we estimated the difference in latitude between the Apuseni Mountains immediately after the emplacement of the Upper Cretaceous magmatics and the reference site selected in the East European Platform. At a 95% confidence level this difference is $14° \pm 13°$ (1540 ± 1430 km, along the meridian). This distance between the two points, which represents the minimum length of the path that the Apuseni Mountains travelled to arrive at the same parallel as the East European Platform, gives the magnitude order of the amount of N–S shortening in the Carpathians. However, the constraint provided by the paleomagnetic data is not very strong because of the large confidence circle of the European pole during the Late Cretaceous.

The paleomagnetic results for Cretaceous and younger rocks from the Pannonian Basin and ad-

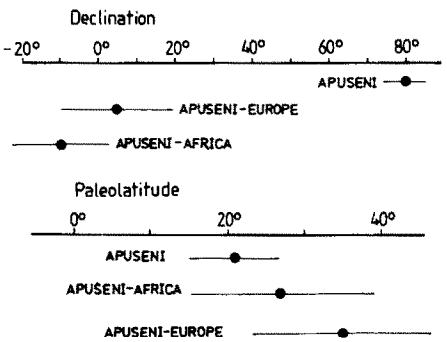


Fig. 7. Theoretical and measured declinations and paleolatitude for the Apuseni Mountains, after Table 3.

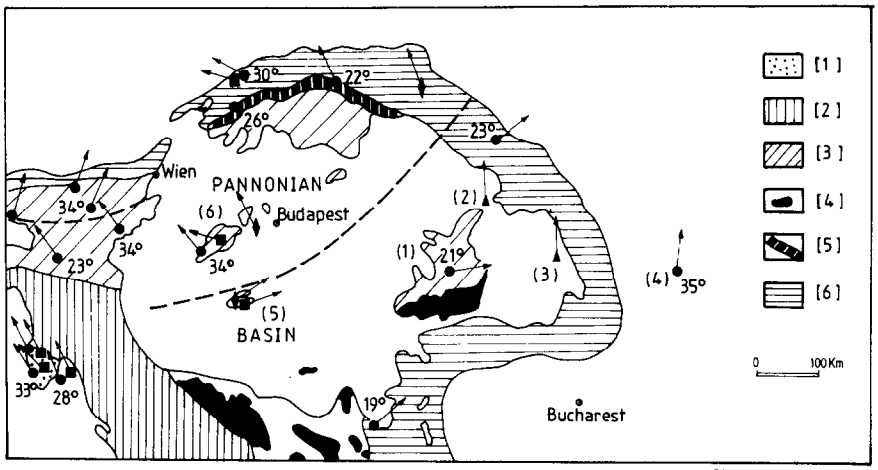


Fig. 8. The horizontal projection of the direction of the remanent magnetizations of the Cretaceous or younger rocks from the Pannonian Basin and adjacent areas. Symbols: squares = Early Cretaceous; circles = Late Cretaceous; diamonds = Paleogene; triangles = Neogene; dashed arrow = theoretical direction of East European Platform (from Table 3); dashed lines = limits between areas with different rotations taken from Márton (1987). The paleolatitude, calculated from Marton's table 1 (1987), is represented by numbers near sites, for the Upper Cretaceous rocks only. Numbers in parenthesis: *1* = Apuseni Mts.; *2* = Oaş-Gutîi-Tibleş Mts.; *3* = Călimani-Gurghiu-Harghita Mts.; *4* = East European Platform; *5* = Mecsek Mts.; *6* = Transdanubian Central Mts. Numbers in brackets: *1* = African foreland; *2* = Southern Tethyan continental margin; *3* = Austroalpine domain; *4* = oceanic remnants (ophiolites) including continental elements; *5* = Pieniny Klippen belt; *6* = Northern Tethyan continental margin. Romanian results are after Pătraşcu (1976), Pătraşcu (in prep.) and this study, the other results are after Márton's review (1987). The geological map is modified after Channel et al. (1979).

jacent areas (Fig. 8) have allowed us to draw some conclusions. It appears that while the Transdanubian Central Mountains suffered a counterclockwise rotation after the end of the Late Cretaceous, the Apuseni Mountains suffered a clockwise rotation. In addition, at the moment of emplacement of the Upper Cretaceous magmatic rock, the Apuseni Mountains area was situated south of the Transdanubian Central Mountains. The paleomagnetic data have provided sufficient results (the paleolatitude, the magnitude and the sense of rotation angle of the Apuseni Mountains) to allow us to propose the first geophysical model for the Apuseni Mountains. This is in accordance with some geological models for the tectonic evolution of the area, which support the clockwise rotation of the Apuseni Mountains (Boccaleti et al., 1974; Burchfiel, 1980; Balla, 1985, 1987). Our results, together with the results from the Mecsek Mountains (Márton, 1986, 1987), provide substantial evidence for a separation of the post-Cretaceous Pannonian Basin in two independent units, namely the Northwestern unit and the Southeastern unit. Márton's result from the Komló laccolith (19 Ma, Mecsek Mts.; see Fig. 8) suggests that a part of the clockwise rotation of the Apuseni Mountains took place after the emplacement of these magmatites. Balla's geological model (1985, 1987) supposes a more-or-less rigid connection between the Northern Apuseni and Southern Apuseni Mountains, as parts of the Southeastern unit. The results of our study support a post-Cretaceous rigid connection between them.

In comparison to the Upper Cretaceous magmatites of Bulgaria (Nozharov et al., 1982, 1987; Nozharov and Veljović, 1984; McElhinny and Cowly, 1980) and of Eastern Serbia (Veljović et al., 1980; Veljović and Nozharov, 1981); those of the Apuseni Mountains were formed farther south. The mean paleolatitude of the Upper Cretaceous magmatites of Bulgaria and Eastern Serbia is around 39° N, but the results in these zones generally have α_{95} greater than 10°. With the exception of part of the magmatites of Eastern Serbia (Veljović et al., 1980), which suffered a clockwise rotation of about 30°, the Upper Creta-

ceous magmatites of Eastern Serbia and of Bulgaria do not seem to have been affected by post-Cretaceous rotation.

The results obtained from the Romanian Neogene magmatites of the Oaş–Gutîi–Tibleş Mountains (Late Miocene age and $D_f = 3°$, $I_f = 64°$ (Pătraşcu, in prep.)) and of the Căliman–Gurghiu–Harghita Mountains (K–Ar age between 7.3–3.9 Ma (Pătraşcu et al., 1972) and $D_f = 6°$, $I_f = 62°$ (Pătraşcu, 1976)) show the cessation of any significant tectonic movement on the border of the Transylvanian Basin after their emplacement. This suggests that the movement of the Apuseni Mountains has not significantly disturbed the above mentioned geological formations for the last 10–12 Ma.

Conclusions

The results of this paper provide the basis for a new tectonophysical model, based on paleomagnetic data, for the post-Cretaceous evolution of the Apuseni Mountains. The main ideas resulting from study of the paleomagnetic data are given below. The area of the post-Cretaceous Apuseni Mountains seems to have behaved as a single block. However, after the emplacement of the Upper Cretaceous magmatites, the movement of the Apuseni Mountains, which up to then had been similar to the movement of Africa, was altered. This was due to interaction with microplates encountered in the route and with the Eurasian plate, which imposed a rotation relative to Europe whose sense and magnitude were different from that of Africa's. The behaviour of the Apuseni Mountains was similar to that identified in other zones included in the tectonic history of the Pannonian basin (Westphal et al., 1986). To reach its present position, the Apuseni Mountains had to move to the north, between the Campanian and, probably, the Late Miocene; during the final period of the movement a large clockwise rotation also took place. Finally, the paleomagnetic results from the Neogene magmatites of the northern and eastern borders of the Transylvanian Basin, suggest that the movement of the Apuseni Mountains more or less ceased 10–12 Ma ago.

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